



Outline

- Inter-observational Distances
- Partition-based Clustering
- Hierarchical Clustering
- Diagnostics and Optimizing Number of Clusters
- Density-based Clustering



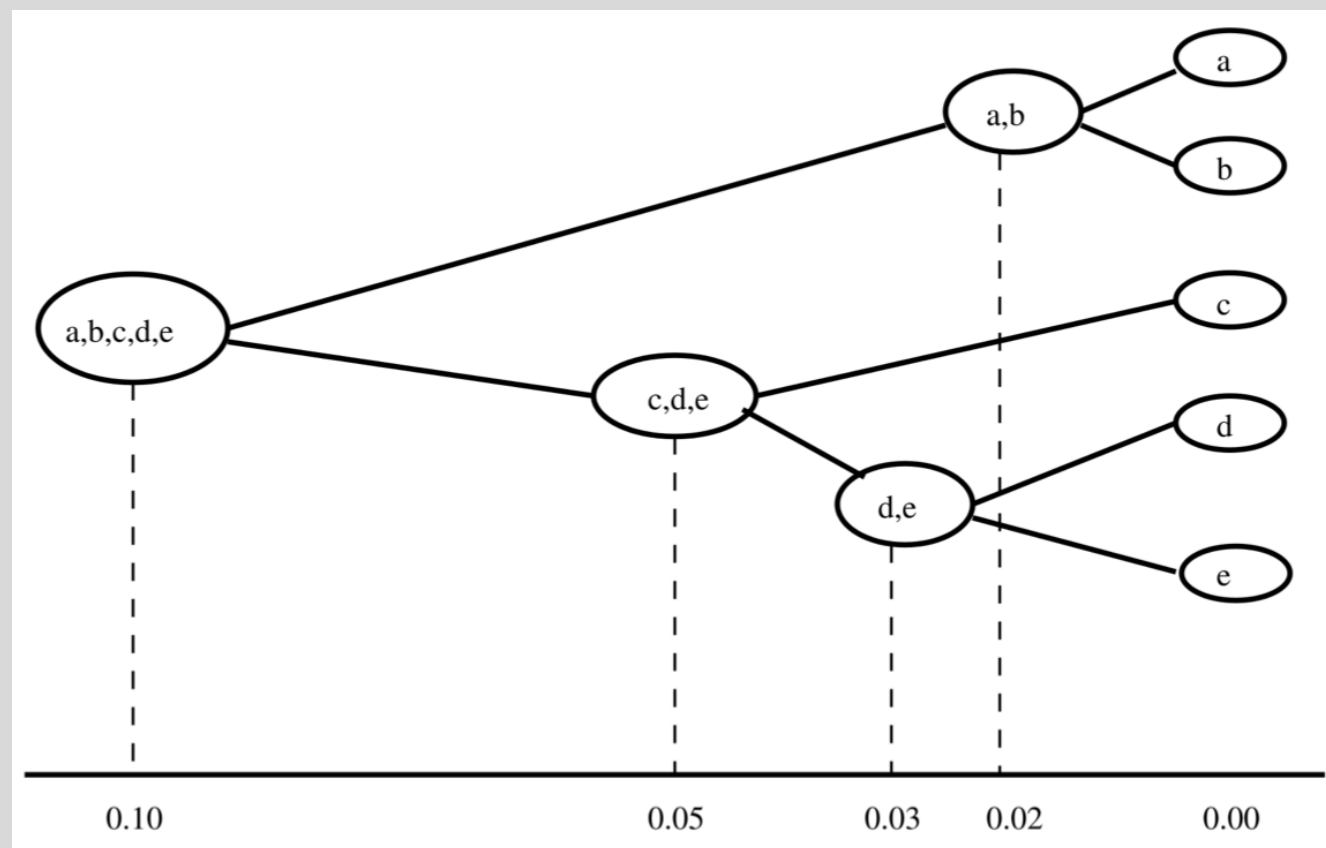
Hierarchical Clustering

- K-Means clustering requires pre-specifying the number of clusters
- Hierarchical clustering does not require committing to a particular number of clusters
- Two types of hierarchical clustering
 - Agglomerative clustering (bottom-up approach)
 - Divisive clustering (top-down approach)



Hierarchical Clustering

Hierarchical clustering can be summarized in a dendrogram





Agglomerative Clustering

The basic algorithm:

- Each observation starts as its own cluster
- At each step of the algorithm, two clusters that are most “similar” are combined into a new larger cluster
- This process of combining clusters is repeated until all observations are members of one single large cluster

Agglomerative clustering is particularly well-suited for identifying small clusters.

Agglomerative Clustering

A few common approaches:

- Complete (or maximum) Linkage Clustering: For two clusters, determine the maximum dissimilarity between any observation in the first cluster and any observation in the second cluster.
- Single Linkage Clustering: For two clusters, determine the minimum dissimilarity between any observation in the first cluster and any observation in the second cluster.
- Average Linkage Clustering: Compute all pairwise dissimilarities between observations in the first and second cluster and calculate the average.

Other aggregation approaches are also possible.



Agglomerative Clustering

A popular approach is known as Ward's Method:

- Instead of joining two clusters based on their distances, use information about the variance of observations within clusters.
- Specifically, at each step, join two clusters whose merged cluster has the smallest within-cluster sum of squared distances.



Agglomerative Clustering – Ward's Method

